Towards Civic AI?

Examining the potentials for civic learning systems for healthier communities

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January 2025



RMIT UNIVERSITY

ABOUT THIS PAPER

This position paper was written by Dr Sarah Barns as part of a Vice-Chancellor's Senior Research Fellowship supported by RMIT University through the Centre for Urban Research, School of Global, Urban and Social Studies, College of Design and Social Context.

ACKNOWLEDGEMENTS

The author would like to acknowledge the support of Stephen Johnston and Professor Iain Buchan, Civic Health Innovation Labs, University of Liverpool.

SUGGESTED CITATION

Barns, S. (2025). *Towards Civic AI? Examining the potentials for civic learning systems for healthier communities* [Position paper]. Civic Interplay, RMIT University.

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1. Context

A novel context for learning

We are living at a major turning point in history. Centuries of knowledge have been integrated into machine learning ecosystems and training models, creating new forms of 'inorganic intelligence' built using sophisticated methods for pattern recognition and recombination. Interfaces for knowledge creation are no longer confined to those created and built by humans; nor are the objects and artefacts of knowledge creation necessarily the work of humans.

At a very high level, we might understand AI as a set of operating procedures and protocols around the use and governance of information, towards its autonomous or 'artificial' replication, as a service or tool. The integration of such AI-based tools has the potential to significantly disrupt institutional workflows, knowledge creation and learning processes, business intelligence and value-sharing relationships as a whole. What kinds of learning systems are we creating here?

Much of the training of large-scale machine learning algorithms has, to date, been led by private technology companies with access to large volumes of data available to them via their business models. 'Personalisation at scale' is one of the key consequences, as finely-tuned recommendation engines adapt and respond to day-to-day actions, intentions and interactions, translated into data points across billions of digital connections. This has, at times, resulted in the tragic consequences of engagement-optimized algorithms stoking outrage, division, and hatred.

Governments and public institutions have contributed to proprietary learning ecosystems by making available large volumes of open data as source materials for large scale training models, with the expectation that the private sector will lead the delivery of digital services utilising this open data. This approach to the 'opening up' of public data and technology capability to market-based competition and innovation, advanced from the 2000s, proved exceptionally successful in facilitating new digital value chains, including initiatives like Google Maps, which is fundamentally dependent on geospatial data provided largely by public agencies.

In this information ecosystem, public authorities acting on behalf of citizens collect standardised information as the basis for infrastructural services provision, and then make that information digitally available to the market to spur wider innovation. But there are significant downsides to this model, requiring a careful rethink.

While making government data widely available, government agencies have at the same time become increasingly dependent on the information services delivered by technology companies. External companies now govern the terms through which information (which was intended to be freely available) is now orchestrated as an input into high-yield value chains, translated into the services upon which we increasingly and routinely depend.

There is, implicit here, a civic learning model or 'program logic', which sees wider participation in data access and use as a means to enabling and accelerating novel, value-generating innovations and capabilities from which we can all benefit. Likewise, there are implicit values imposed here around the protection of personal data, ensuring minimal linking of citizen data records comparative to the personalisation algorithms widely adopted in the commercial data sphere.

As has become evident, however, this learning model has ultimately failed to 'close the loop' between processes of value exchange that occur between value-creation and value-extraction. Beneficiaries of government open data have been obligation-free, even as they reap the benefits of free information.

Much of the critical commentary attached to the ills and failings of the digital economy we live in today has centred on the problems attached to data surveillance. Surveillance critics highlight the extent to which data pertaining to personal actions and behaviours is commodified via advertising based business models, which target and seek to influence personal preferences and behaviours via recommendation algorithms. Privacy is traded in exchange for access to sophisticated, but basically free, digital products.

As troubling as this situation may be, the surveillance critique, which calls for greater protections of personal privacy in data exchanges, says little about the lost value to society when its public data, and the public funding of its research institutions, is given away to privatised data ecosystems. This critique can inadvertently protect the opacity of proprietary algorithms, on the basis that they protect underlying 'private' data.

Beneficiaries of government open data have been obligation-free, even as they reap the benefits of free information.

Towards new learning ecosystems?

We need another way. We should recognise that our options today are not limited to either a) the public monopolisation of big data or b) private sector monopolisation via proprietary algorithms. Citizens and educators are in a position to advocate for alternate models, and should work collaboratively, at scale, to test and implement possible scenarios and alternatives.

In this context, educational institutions can play a key role. For institutions seeking to build learning capabilities as a fundamental mission or purpose, now is the time to carefully consider the means by which their informational outputs are being shared and distributed. How are they being put to use?

Conditions for generative learning include not only teaching and research outputs themselves, but the data-generating activities of wider institutional activities, from capital works programs to public engagement programs. It is no longer the case that institutional actors can be neutral in relation to their ecosystem inputs.

What the generative AI explosion has demonstrated, with acute clarity, is that the conditions for machine learning to operate at scale are, today, critical to the operational and governance capabilities of an organisation vis-a-vis its overall mission. To thrive in an era advancing the applications of generative AI, organisations today must become razor focused on aligning and optimising their informational ecosystems - their 'tech stack' - around mission and purpose.

It is no longer sufficient to 'make information available' if there is not a concomitant investment in the learning context that makes use of this information - that is, the informational ecosystems and network relationships that govern and manage its application. This learning environment is the means by which information is put to use: in other words, its data value chain.

This is because leveraging the benefits of AI for an organisation means engineering informational workflows in ways that complement and extend existing organisational capabilities, while addressing gaps and areas of weakness. Much workflow optimisation currently leverages the capabilities of personalisation engines and customer relationship management (CRM) software, reflecting an inherent bias towards conditions of profit maximisation from all digitally-mediated relationships.

However, it would be wrong to reduce the capabilities of co-intelligent information ecosystems to that of sales engines alone. Institutions and non-profit organisations whose relationships are centred around 'training' and enabling different kinds of relational ethics can also be explicitly focused on how they are leveraging co-intelligence towards different ethical goals and outcomes.

For institutions whose legitimacy rests upon their ability to serve the wider public, it is very much the time to make data governance an institutional priority. Key accountabilities and 'missions' should directly inform how information flows are being managed. This means not only prioritising the security and privacy of institutional data, but also ensuring the wider corpus of information being generated through the activities of those who contribute to its mission is valued as an input to co-intelligent, AI-based training models.

For institutions whose legitimacy rests upon their ability to serve the wider public, it is very much the time to make data governance an institutional priority

This should then inform the question of whether existing ecosystems are aligned to institutional mission. Are the training models in place contributing to the citizenship goals of the institution? Or are they accelerating the learning ecosystems of a profit maximising platform upon which the institution is increasingly dependent, and whose mission and ethical frameworks may in fact be completely at odds with those of the institution?

Rebuilding information architectures of public and mission-focused organisations, in ways that optimise the benefits of co-intelligence, will not be easy. The work requires a careful re-consideration of what it means to act digitally as a citizen, an educator, an environmentalist, a health care practitioner in a data intensive ecosystem that seeks to achieve impact. There is the potential here to establish learning ecosystems around practices of active citizenship, democratic accountabilities, public benefit and environmental stewardship. As Rachel Coldicutt aptly put it, our goal should be to "make AI work for 8 billion people not 8 billionaires" (Coldicutt, 2023).

Now is the time to lean in, and try something different.

2. Civic AI as a learning model

If the case for change is now clear, the methods are far from certain. This is why experimentation is critical, informed by an explicitly values-based approach.

From a governance perspective, the health of democracies depends on the contributions of civil society as a 'third sector', forming an essential part of the checks and balances between public and private interests in a stable and effective democracy. Civil society engagement is part of a society's culture of participation, which ultimately enhances the democratic nature of decision-making by state authorities. Existing in the gap between the state and the private sector, civil society can be, as a whole, amorphous, spanning families, churches, non-profit organisations, environmental campaigners, sporting clubs and so forth. Ultimately, the role and value of this 'third sector' in a democracy lies in the capabilities formed by people acting collaboratively with a shared interest or vision of public interest.

The domain of civil society represents an important area of attention in our era of AI, as it slides towards techno-nationalism and extractive, profit-maximising data surveillance. Currently, the information ecosystems enabling civil society organisations to 'advocate' for commons-based principles and outcomes remain highly fragmented. Were these to be strengthened, non-profit groups, educational institutions, citizens and mission-led organisations could act more autonomously and judiciously to inform how their 'data assets' are used within wider value-chains of decision making, service provision and profit maximisation.

As should now be clear, data alone is not necessarily an asset: it only becomes valuable when there is a learning system or ecosystem established to define its relative value.

To do this, however, the value of civil society organisations would need to be considered for their role within algorithmic learning systems, which operate according to defined logic models that grow and improve over time with more training data. As should now be clear, data alone is not necessarily an asset: it only becomes valuable when there is a learning system or ecosystem established to define its relative value.

Civic learning capabilities for civic AI?

Civic AI provides a method for redesigning distributed learning ecosystems and workflows in ways that evidence the contributions of civil society to the achievement of public interest outcomes. It aims to 'close the loop' between civic, non-profit activities and their benefits, by creating the necessary learning models that demonstrate and evidence value-generation over time.

The value of civic AI lies in the data ecosystems it helps to activate, linking citizen and civil society activities with their missions or goals. Many civil society organisations define themselves as 'mission-driven' and 'for purpose' organisations, and have established frameworks in place to measure and define the value of their activities towards these goals. The test is to activate these information ecosystems as workflows in support of wider missions.

Civic AI builds on the existing citizen science movement, which has demonstrated clear benefits to activating citizens as data collectors in service of specific scientific or environmental monitoring goals. Citizen science methodologies can be limited, however, by the adoption of bespoke apps or data collection methods that require active participation by contributors.

There is a relatively limited pool of data collected through citizen science methods compared to the volumes of data collected by proprietary algorithms over long time periods. Addressing this imbalance in ways that reflect the values of citizenship and democracy is a critical challenge. Do the methods of data surveillance advanced through proprietary algorithms justify their expansion into civil society domains?

The answer is no. However, it is imperative that alternate learning ecosystems for citizen and civil society data be established, with appropriate checks and balances in place around how personal data can be shared and used in the service of civic AI. To achieve this, citizens need to have the ability to make informed choices about how their data can be used to support public value missions, rather than being utilised purely within private data ecosystems.

Civic AI naturally builds on the existing citizen science movement, which has demonstrated clear benefits to activating citizens as data collectors in service of specific scientific or environmental monitoring goals.

The following section explores current and potential case studies for the development of civic AI as a learning system that operates outside of proprietary algorithms.

3. Case studies

I. Civic Health Innovation Labs, University of Liverpool

Health system learning is a domain significantly disrupted by the rise of consumer-facing interfaces for biometric tracking and other data sensing devices adopted by citizens to track their own health. While this creates a 'data deluge' on the one hand, institutional contexts for public health care provide limited capabilities for citizens and health care providers to link data generated by citizens opting for biometric sensing apps to wider health care services, due to well established protocols around data privacy and data sharing. Clinical trials operate with far more limited data as compared to the data deluge resulting from the explosion in health care apps and interfaces directly available to consumers via their app store. In this context, commercial healthcare applications have been able to accelerate significantly in scope and capability, while public health care trials continue with more limited clinical data.

In this context, the University of Liverpool's Civic Health Innovation Labs (CHIL) is taking active steps to design novel data sharing principles, practices and relationships that underpin the work of social care towards a civic AI health learning ecosystem.

Led by Professor Iain Buchan, a public health physician and data scientist with experience working on AI and health care services, CHIL advocates for novel 'co-production' and data sharing models to inform how patient data is being used to optimise health care.

The concept of 'health system learning' is adopted here as an ecosystem lens to understand characteristics of data use across health services, incorporating a critical mass of linked data along with the methods and expertise needed for health systems to adapt to the needs of the populations they serve.

The CHIL program seeks to address a gap in data availability and capability opening up across public and private data ecosystems - what they call a 'health data paradox'. With a focus on health informatics, CHIL advocates for the linking and scaling up of connected health data towards greater overall health system learning. As Ainsworth & Buchan state (2015, p. 482):

> *We are on the cusp of a data deluge, where individual data from bio-medical sensors, GPS, and accelerometers can be combined to with population and environmental data to provide a rich longitudinal picture of our health.*

Addressing the health data paradox, the authors advocate for 'co-production' as a method, where data generated by patients can be used to optimise care. This model requires the interaction of patients, clinicians and machine algorithms in order to "extract the signal from the data and to transform it into knowledge for shared decision-making" (Ainsworth & Buchan, 2015, p. 482). Their proposal sees health information systems moving from separate requirements for machine-clinician and machine-patient interaction towards a more integrated triangle of digital health. This is pictured below.

Figure 1. Digital health triangle. Ainsworth and Buchan (2015, p. 482). The figure can be seen in the original paper.

Within the triangle is the concept of 'information / avatar', acting as a synthetic representation of patient/citizen health records to inform decision making. As the authors note, communication within this triangle will require the union of records and models that represent an individual's health to be conveyed

with context aware 'personalities'. They describe these personalities as being "more avatar than filing cabinet (the stale paradigm of medical records)."

The concept of a patient or 'health avatars' plays a critical role here, acting as 'social machines' that emit early warnings of conditions deteriorating in community settings, triggering actions that might, for example, avoid hospital admissions.

A key driver of the 'digital health triangle' is the broader crisis in public health care provision afflicting many communities across the United Kingdom. Current models for health service provision under the umbrella National Health Service (NHS) do not match complex demands for its services, including the capital requirements to maintain safe care environments. (Wickens, 2024). A shift away from hospital care towards community based health care has been advocated for many years, but there are systemic factors preventing the re-localisation of health care (Baird et al., 2024). In a recent review, Baird et al (2024) concluded:

> *The failure to grow and invest in primary and community health and care services ranks as one of the most significant and long-running failures of policy and implementation in the NHS and social care for more than 30 years. If this shift in focus does not happen, more expensive hospitals will need to be built to manage people with acute needs that could have been prevented or better managed.*

While care in social and community settings goes under-resourced, spending is becoming more heavily weighted towards the acute hospital sector. The review describes a 'cycle of invisibility' existing for primary and community health and care services, which are "hard to quantify and easy to overlook". This helps to reinforce existing 'hierarchies of care' that see urgent problems taking priority over longer-term issues, such as services that can prevent the development or escalation of health care problems that surface in acute care settings. From the training of staff to the financial structures underpinning approaches to return on investment, policies and strategies are not aligned with the vision of care focused on communities (Baird et al., 2024, p. 2).

A novel approach to managing citizen data represents a key intervention in this 'cycle of invisibility', which sees public health care resources remain overly skewed towards acute care settings. While not stated explicitly, the weighting of public health care resources towards acute care settings can be interpreted as an unfortunate product of the current public health data ecosystem. For example, a focus on 'patients' as the beneficiaries of care within existing public health care settings, as opposed to 'citizens', necessarily skews data towards those already active in the existing health care systems, with preventative contexts receiving lesser weighting and comparatively less visibility.

Instead, through closer interactions with health information, citizens and health care providers are better equipped to target preventative healthcare, and savings generated through greater investment in community and social determinants of health more effectively captured and evidenced over time.

CHIL has developed a data schema in response to these challenges, which sets out a 'connected city model for health system learning'. The concept responds to the urgent requirement for public health care to build capabilities in social and community care settings, including investment in preventative health care.

Figure 2. A Connected City Model for Health System Learning. Ainsworth and Buchan (2015, p. 484). The figure can be seen in the original paper.

The initial scheme presented here was developed in 2015, predating by some years the generative AI wave launched when OpenAI launched its consumer-facing ChatGPT (OpenAI, 2022). Emerging from the discipline of health informatics, the scheme for the Connected City Model highlights how novel data configurations and relationships between citizens, health providers, researchers, policy makers and industry could begin to operate at scale.

The central axis of the scheme is centred on the concept of the 'Ark', incorporating involved citizens, data managers, public health analysts and so forth. Members of this 'ark' are essentially custodians of data, given permission by data owners to integrate data into insights from which new tools and services and applications can be built.

Key concepts developed by Ainsworth and Buchan (2015) include:

- Pipelines of evidence: Each evidence pipeline represents a bond of trust between the generator of data and their consumer. Such 'evidence pipelines' are commonly adopted by consumers in their adoption of healthcare apps, but what does that look like if the same 'evidence pipeline' was then extended into a healthcare setting? In this instance, a pipeline has a 'custodian' who represents the patient/citizen and oversees the integration of their data into wider clinical settings.
- Temporary permissions: The concept of 'evidence pipelines' depends upon temporary permissions being granted between data owner and data user, and the custodian or intermediary of this exchange. Permissions are granted in the form of a licence issued for a defined period of time, after which time participating parties have the power to either revoke or renew the licence.
- Cross-sector innovation and value-creation: The connected city model for health system learning also prioritises cross-sector innovation. Investments in community infrastructure that support preventative health care, reducing demand for acute hospital care, are captured as health system learning benefits. This enables continuous quality improvements across government agencies. At the population level, using civic AI to help target resources would encourage the system-wide maximisation of value.

The CHIL model has advanced in recent years, through creation of a Civic Data Cooperative (CDC), which treats citizens not as passive sources of data, but as active stewards of their local, digital health information (University of Liverpool, n.d.-b). Through initiatives like "What's Your Problem?" and "Round 'Ere" (Civic Data Cooperative, n.d.-a, n.d.-b), residents collaborate directly with researchers and healthcare providers to address local health challenges. The CDC's approach represents a fundamental shift in thinking about health data governance. Rather than treating personal health information as a commodity to be extracted and exploited, it views it as a common resource to be managed for collective benefit.

II. Civic AI activation zones: a model for healthy learning systems?

The CHIL model introduces novel concepts around data and AI use for health system learning. It creates opportunities for citizens to work in partnership with health care authorities and researchers, to develop greater autonomy and capability in how their data might be utilised within big data learning models.

Public health care is an instructive example, because it links highly-personal, biometric information with public services. Establishing guardrails - like temporary permissions and licenses to use - are critical here. Health care is also an area facing increasing budgetary and cost increases associated with ageing populations and the negative externalities associated with wealth inequality and climate change.

A ‘joined up’ approach to health, linking place-based strategies with health outcomes, has long been advocated, but there is little evidence of success. Cross-sector innovation of this kind requires inter-agency, cross-sector collaboration, across distinct institutional sectors from urban design, landscape planning, architecture, and health services. More joined-up data models - learning systems - are crucial.

Innovation precincts represent a well-established method for co-ordinating place-based developments through clustering and novel governance mechanisms. Tech-based innovation precincts and business parks are well known, and health innovation precincts are an opportunity to test and scale new learning models across different co-located institutional actors.

Leveraging the model of the innovation precinct, there is an opportunity to extend the innovation precinct concept, to create ‘civic AI activation zones’ for testing and evaluating new civic learning models. The advantage with a place-based approach lies in the integration of linked data that span diverse community-based contexts for learning.

What could this entail?

In the first instance, a civic AI activation zone could test ways to give ownership, control and shared value of data as much as possible to the individual, their network of family and friends and local communities. Citizens become active stakeholders rather than passive subjects, with data sharing clearly linked to wider social benefit outcomes.

The approach builds on a number of citizen-first initiatives led by city governments in recent years. One of the most notable of these was the ‘New Data Deal’ led by the administration of Mayor Ada Colau in Barcelona from 2016 (Fernandez-Monge et al., 2024; Monge et al., 2022). The City revised existing ‘smart city’ contracts with technology providers to ensure the city had greater access to the data previously locked up in private contracts. The City also led an innovative initiative called ‘DECODE’ which introduced privacy preserving encryption methods to citizen-data sharing in local precincts, focused around key city planning objectives (Sagarra et al., 2019). The DECODE initiative attracted much attention, at the forefront of urban policy initiatives embracing more active methods for ‘data custodianship’ by citizens, however ultimately did not extend beyond the prototype phase.

Civic AI learning systems build on the principles established by city governments, like the City of Barcelona and also the CHIL Civic Data Cooperative. However, the ambition is to ensure a greater volume of data available via contemporary sensing devices is incorporated across more longitudinal time frames, in ways that directly align with mission-led goals and objectives. This takes advantage of new ambient computing interfaces that incorporate multi-modal sensing and AI agents, thus opening up more ‘ambient’ and autonomous methods for data collection and sharing.

Civic AI activation zones can also test new methods for incentivising learning systems. Incentives could be in the form of sponsored trials and partnerships, or testing Web 3 integrations, which translate data into tokens with value. Sponsored partnerships could enable coalitions of mission-driven civil society organisations, citizens and partners to collaborate to unlock key insights, addressing data gaps needed to spur collaborative investment. This work would require a collective, collaborative focus on the design of active indicators and learning loops reflective of mission-led priorities. This would leverage distributed and federated intelligence through cross-community knowledge sharing, agreements on standards and data / insights pooling.

A critical gap to date in the advance of citizen-led data sharing initiative has been the issue of scale. While private data algorithms are able to work across global scales to train recommendation engines and build algorithmic insights, many civic-led initiatives remain highly localised. As a consequence, data volume is lower, and focus areas become centred on highly localised topics. However, through novel collaborations and partnerships, Civic AI learning ecosystems can also operate at scale, through a 'mirrored sandbox' approach. With common taxonomies and evaluation models, aided by AI-assisted data processing, localised initiatives can scale up.

Think of this as creating teams of humans and AI working together to strengthen social fabric. While humans bring context, wisdom, and lived experience, AI can help process vast amounts of community input, identify patterns of agreement, and suggest areas for collaborative action. This isn't about replacing human connection, but augmenting it with tools that help us translate data into insights more quickly and effectively.

Universities play a critical role in nurturing and fostering these novel learning ecosystems. Working across stakeholder networks, civil society organisations, students, start-ups and researchers, trials of Civic AI activation zones can test collaborative learning models that advance societal missions using AI-based workflows and interfaces. Through parallel experimentation, pattern recognition, and rapid learning loops, communities can scale successful interventions through replication.

MIRRORED SANDBOXES

Mirrored Sandboxes are tools for scale, and contain three parallel tracks: research, demonstrators and field building.

Research engages academic partners to map interventions across communities, building an "Oracle" of what works where and why. Through shared standards and privacy-preserving protocols, it systematically captures patterns of success and failure, enabling communities to learn from each other's experiences without compromising data sovereignty. Each community maintains control of its data and algorithms while contributing patterns of success and failure to a global learning network.

Demonstrations engage socially-minded entrepreneurs to turn insights into action through locally-based prototypes. From Austin's diabetes management to Liverpool's obesity interventions, these living laboratories test novel approaches while attracting world-class innovators in AI, blockchain, and quantum computing to solve real community challenges. Funding is provided by local donors as well as those interested in scaling insights globally.

Field Building engages funders, researchers, marketers, policy makers and the public, to change the narrative around AI and community health. This work attracts funding, promotes policies that support local data sovereignty and engage the public to both care about their health, and who owns it. By creating an international community of practice, comprising experts and communities, it builds resilient networks that can function regardless of national turbulence, enabling lasting systems change.

Conclusion: now is the time to make communities healthy

Local benefits, community connections and shared common resources sound quaintly Victorian; a throwback to a more innocent age, when milk was delivered in glass bottles, supply chains weren't just-in-time, and the prevailing political narrative was more liberal, in every sense. It's at times like this when novel, locally-rooted, non-partisan approaches matter most. And nothing matters more to voters than the health of their loved ones.

The narrative that AI requires massive centralization is compelling but false. Early results from federated networks are promising. The question is whether they can expand fast enough to prevent Big Tech's dominance of AI. As AI capabilities advance exponentially, the window for establishing more equitable models of data governance is closing. For communities typically left behind by technological progress, the stakes could hardly be higher.

Perhaps most critically, the majority of health outcomes are determined by social determinants — where people live, what they eat, how they sleep, and with whom they interact — not by episodic medical interventions. Yet most AI healthcare applications remain stubbornly focused within hospital walls, missing the broader context that drives longevity and well-being.

An alternative approach is emerging: when AI is locally owned and democratically governed, it can be optimized for what communities actually want, such as connection, compassion, understanding, and collective wellbeing.

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